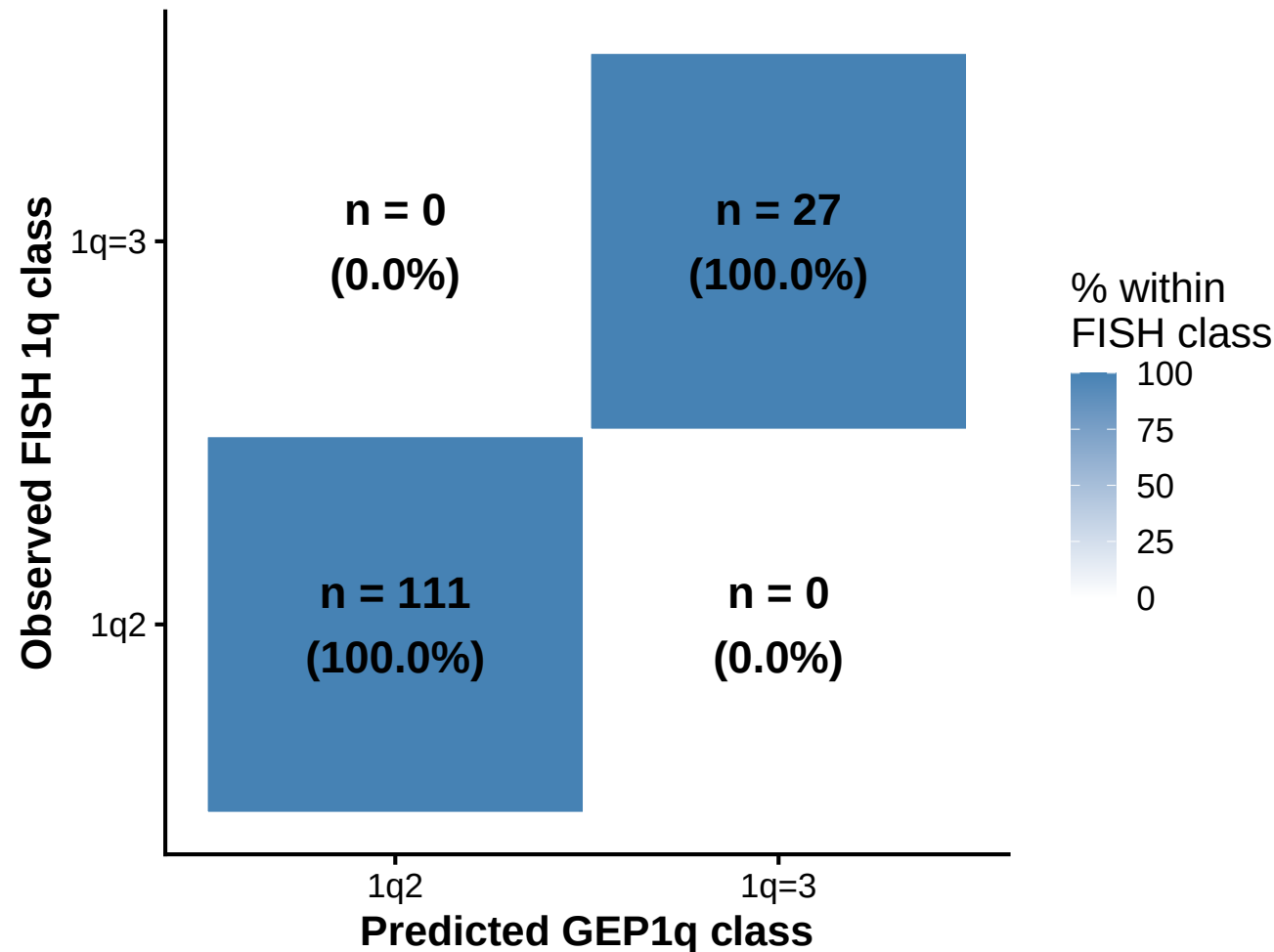


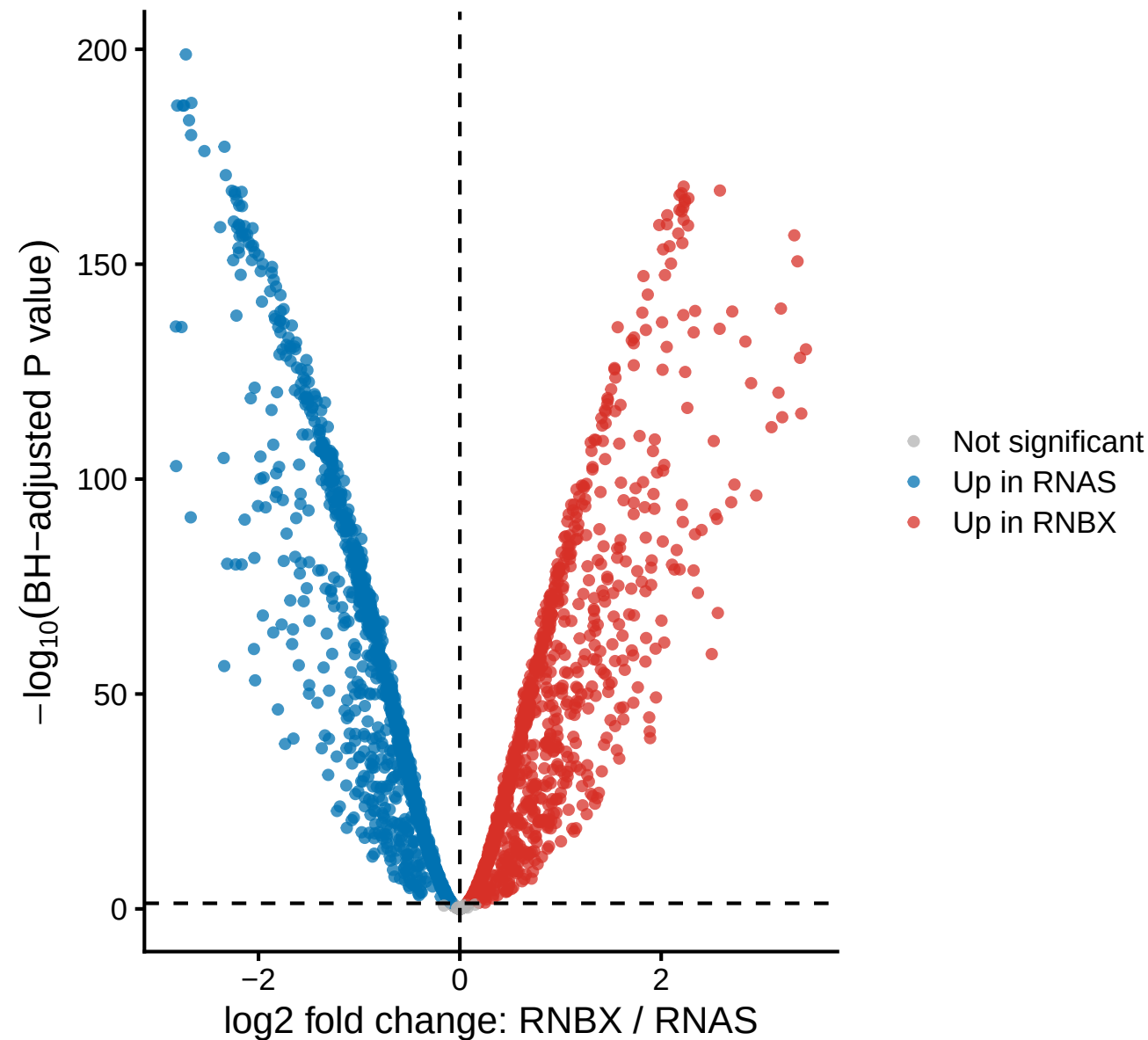
Strict Held-Out GEP1q Classification

Accuracy = 100.0%



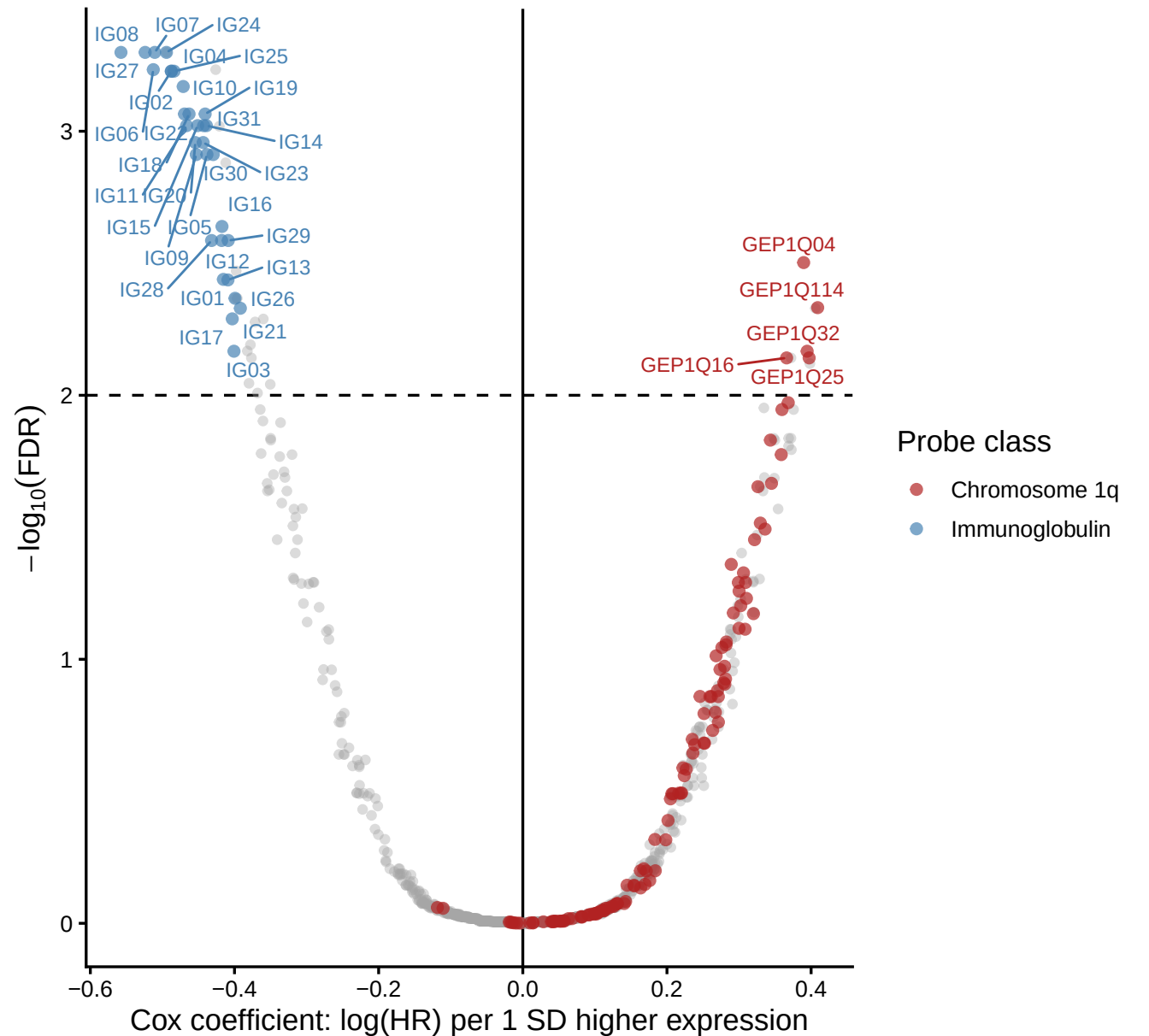
Paired NDMM RNBX vs RNAS

PATID-matched pairs: n = 285



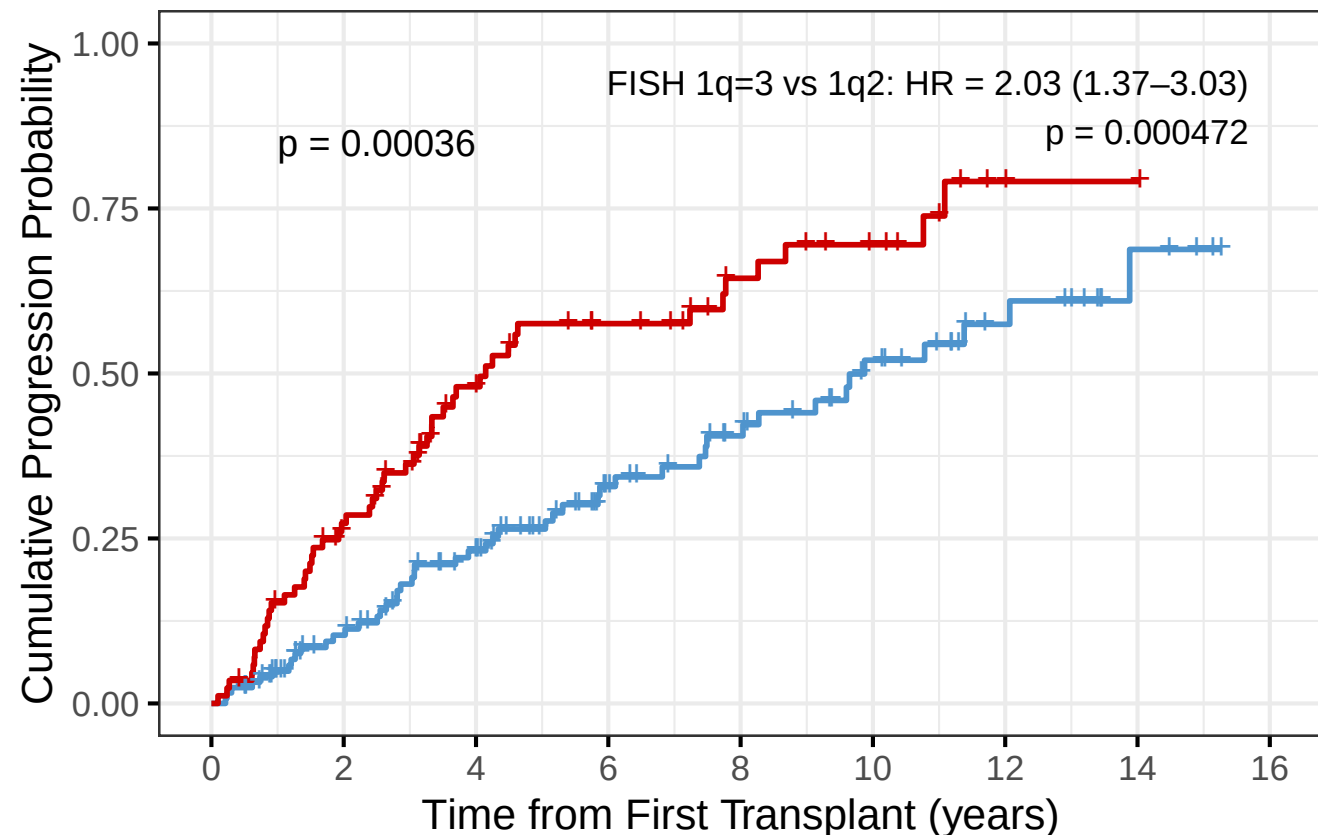
Genes Associated with Time to Progression in GEP70 Standard-P

Baseline CD138+ NDMM; n = 242; events = 112; effect per 1 SD higher expression



Time to Progression by Chromosome 1q (FISH) Copy Standard Risk (GEP70); TT2–TT4 (n = 211)

FISH 1q copy class 1q2 1q=3



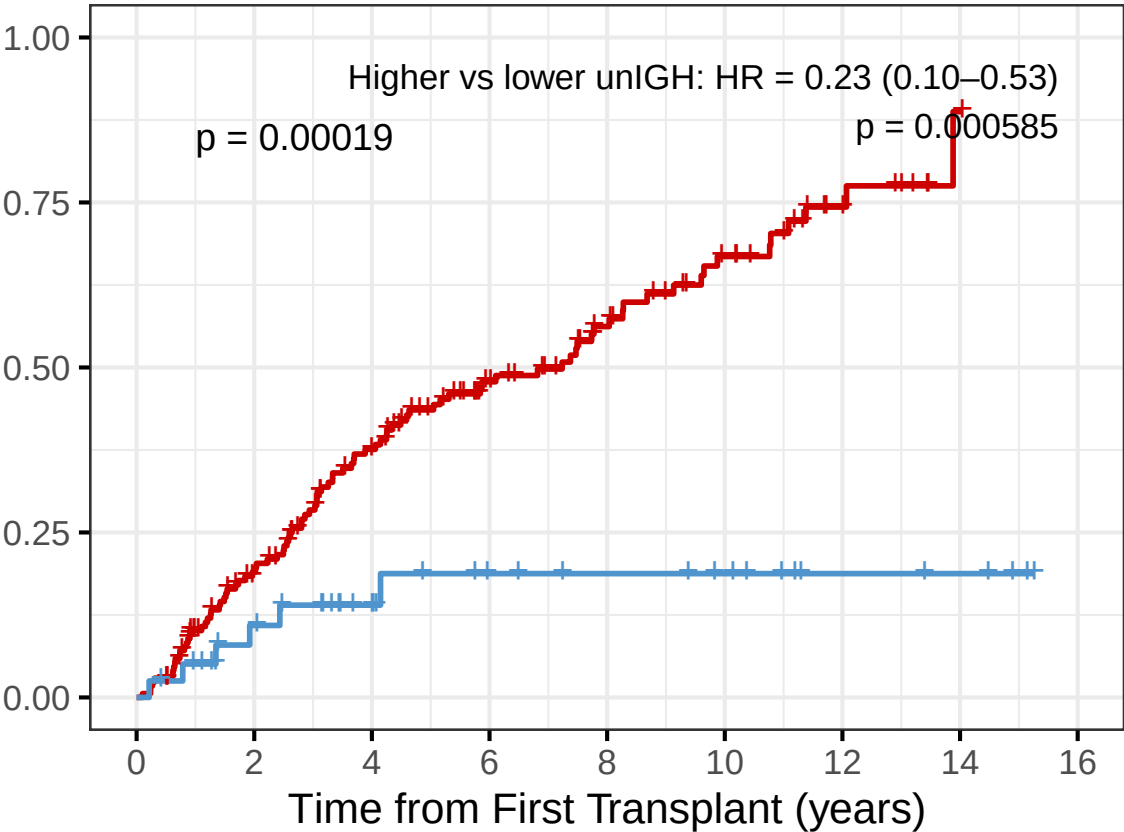
No. at risk

1q2	125	96	73	47	35	23	12	4	0
1q=3	86	58	34	23	14	9	2	1	0

Time to Progress by Uninvolved Immunoglobulin (unIGH) Standard Risk (GEP70); TT2–TT4 (n = 211)

Clinical unIGH + unIGH = 0.32 + unIGH > 0.32

Cumulative Progress Probability



No. at risk

unIGH = 0.32

171	124	86	56	37	22	9	1	0
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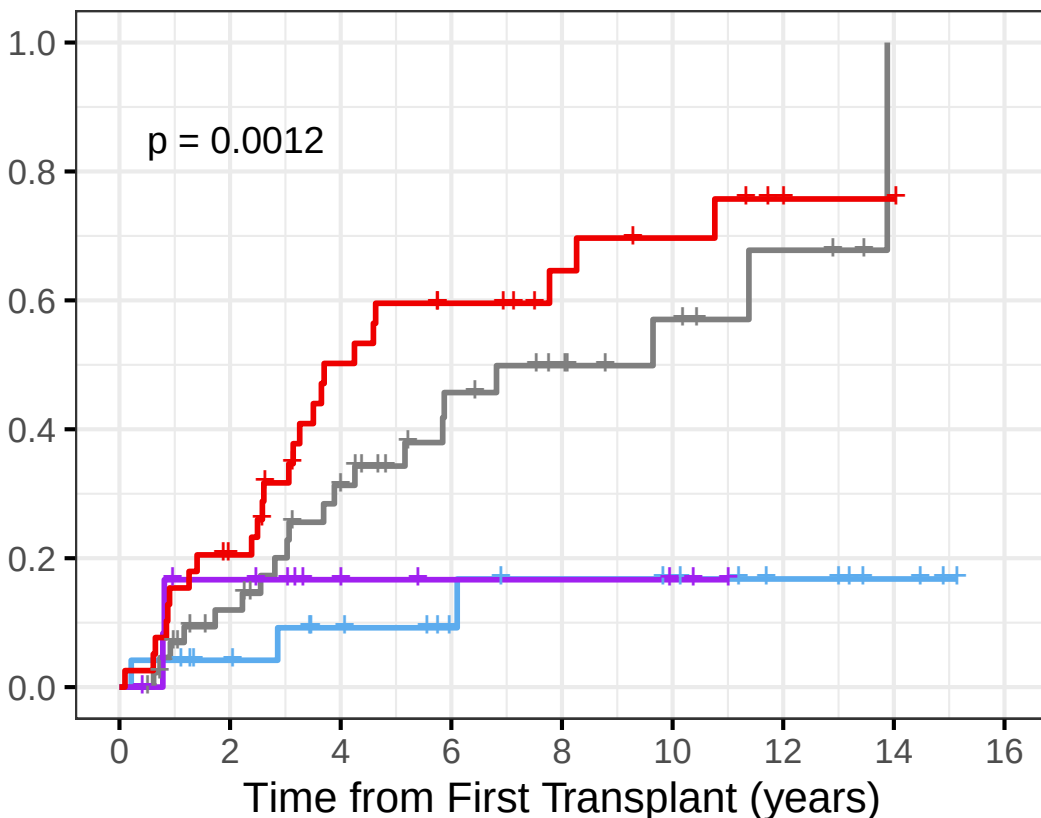
unIGH > 0.32

40	30	21	14	12	10	5	4	0
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Time to Progression by FISH 1q Copy × unIGH

Total Therapy: TT3a, TT3b, TT4-S (n = 121)

Cumulative Progression Probability



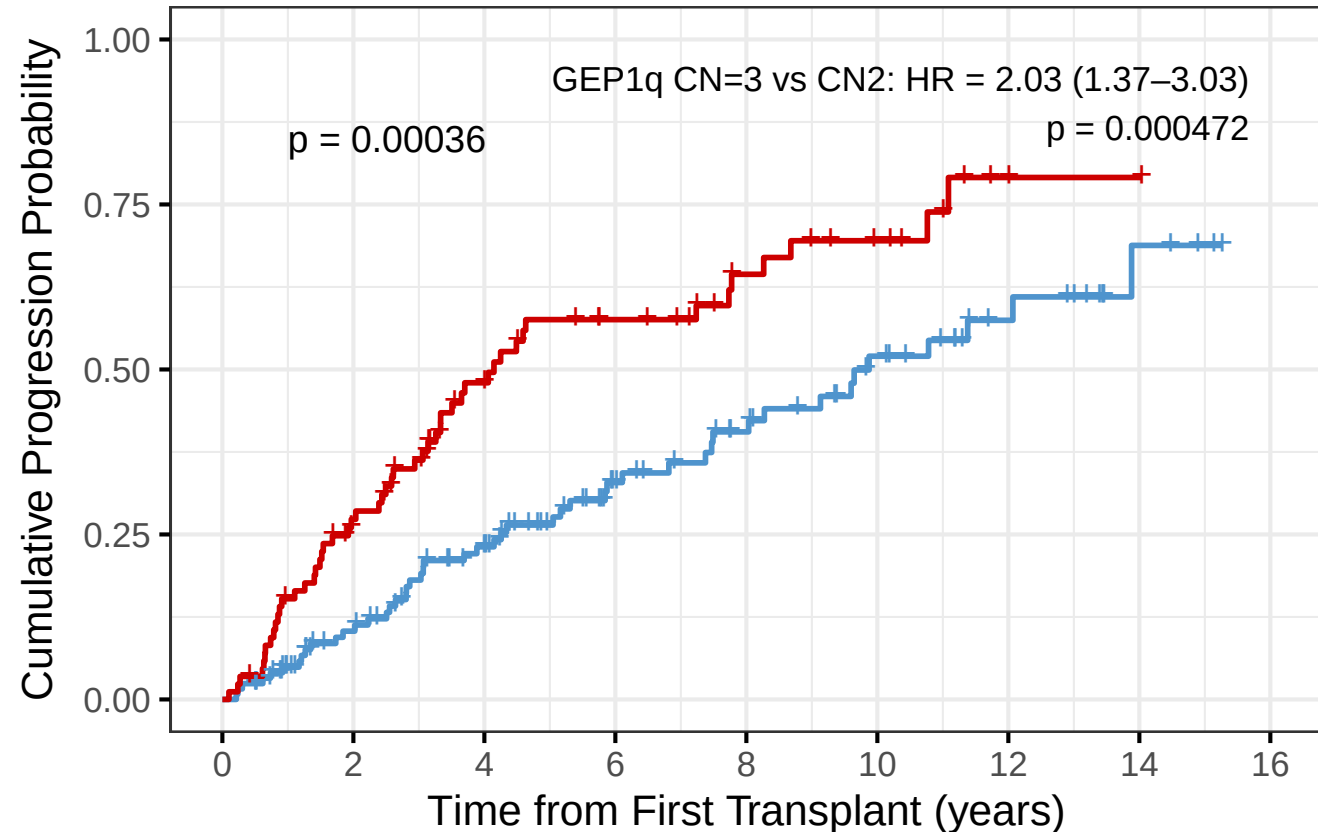
Group + 1q2/unIGH High + 1q2/unIGH Low + 1q=3/unIGH High + 1q=3/unIGH Low

No. at risk

1q2/unIGH High	24	20	16	12	10	9	6	3	0
1q2/unIGH Low	45	34	23	14	10	6	3	0	0
1q=3/unIGH High	13	9	5	3	3	2	0	0	0
1q=3/unIGH Low	39	29	16	11	7	5	2	1	0

Time to Progression by Penalized GEP1q Copy Class Standard Risk (GEP70); TT2–TT4 (n = 211)

GEP1q CN2 CN=3



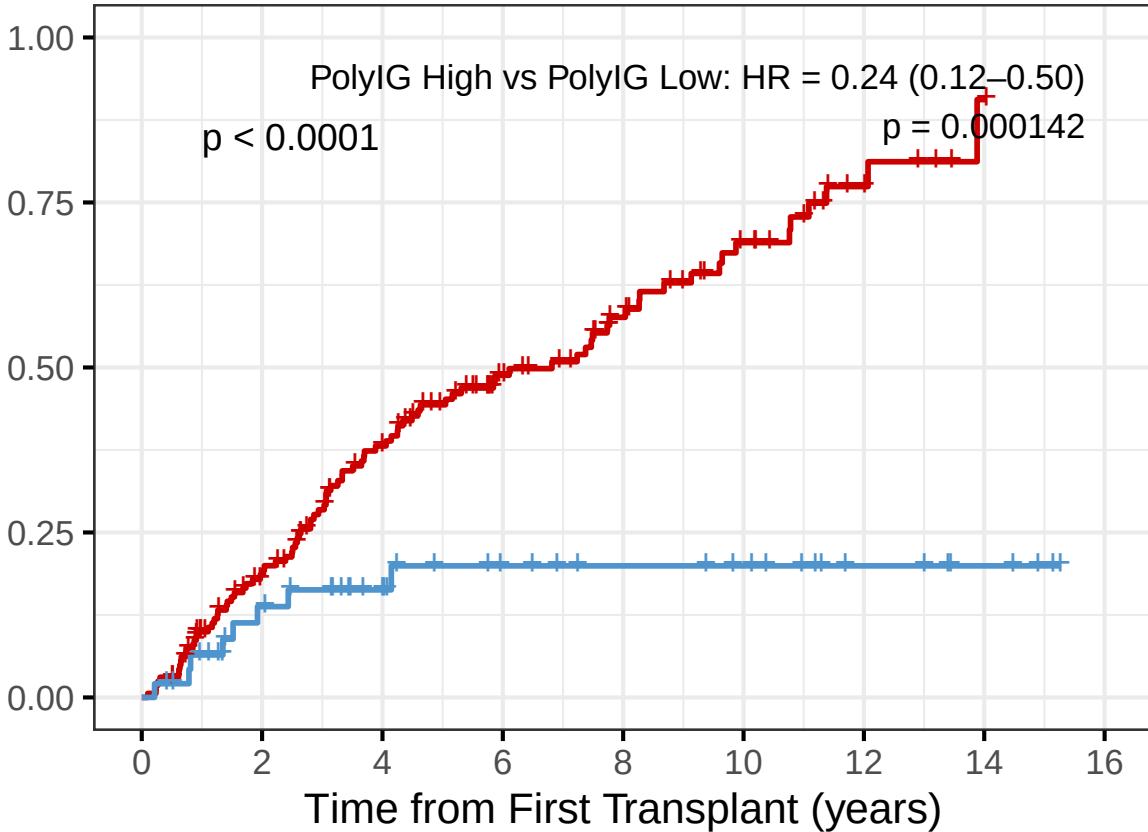
No. at risk

CN2	125	96	73	47	35	23	12	4	0
CN=3	86	58	34	23	14	9	2	1	0

Time to Progression by PolyIG (GEP) Standard Risk (GEP70); TT2–TT4 (n = 211)

PolyIG group + PolyIG Low + PolyIG High

Cumulative Progression Probability



No. at risk

PolyIG Low

163

119

81

52

34

19

7

1

0

PolyIG High

48

35

26

18

15

13

7

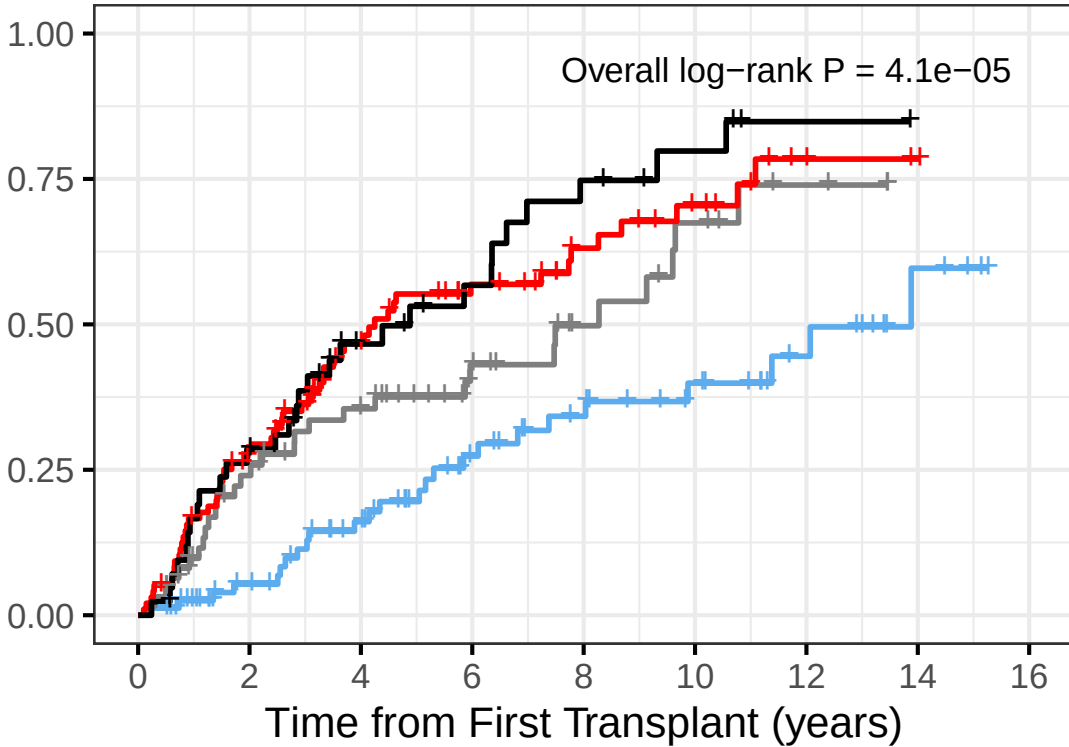
4

0

Time to Progression by Penalized GEP1q Copy × P Standard Risk (GEP70); TT2–TT5 (n = 285)

GEP1q × PolyIG group + CN2/PolyIG High + CN2/PolyIG Low + CN>=3

Cumulative Progression Probability



No. at risk

CN2/PolyIG High

82 65 51 34 26 19 11 4 0

CN2/PolyIG Low

63 42 32 20 12 7 3 0 0

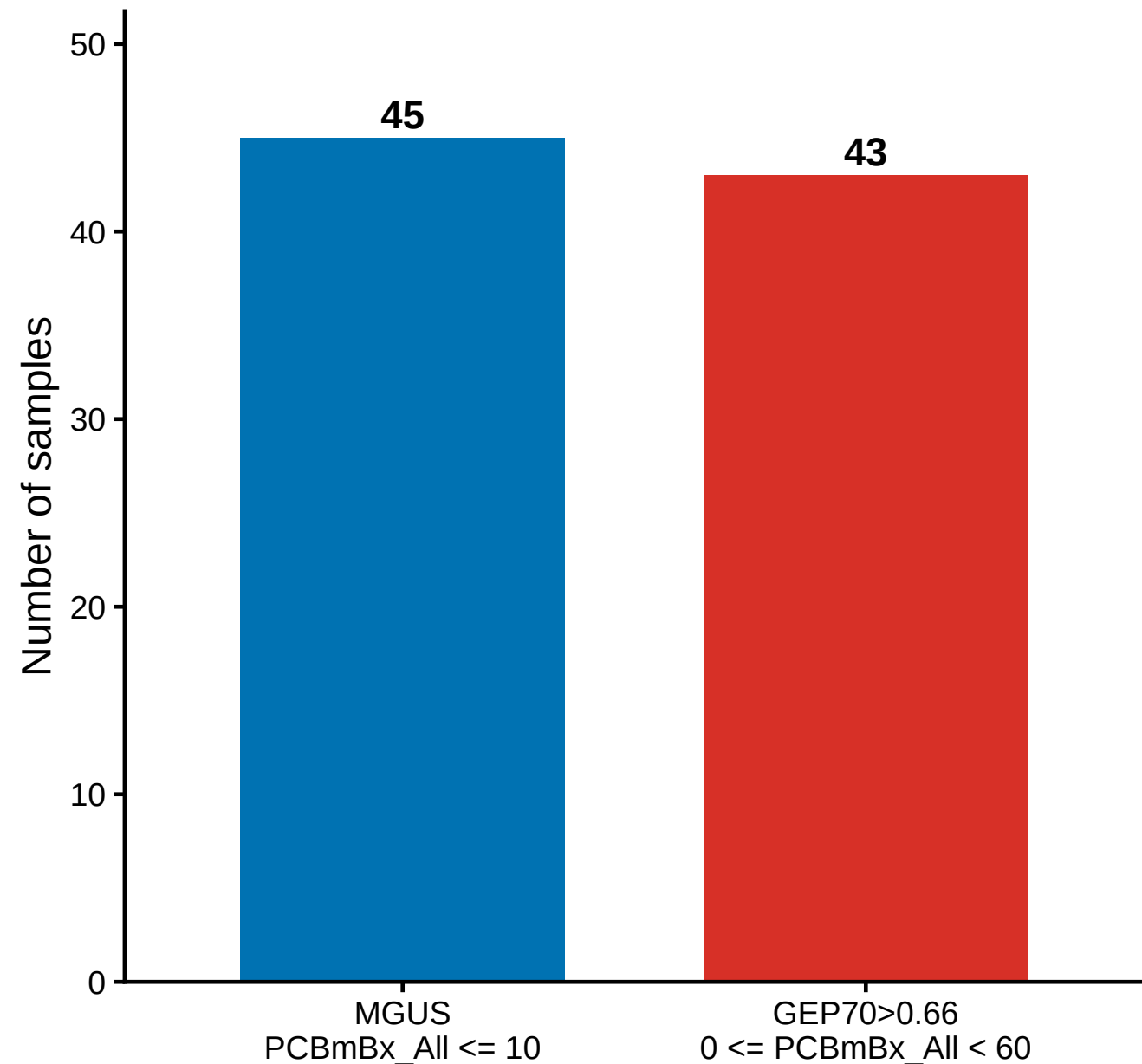
CN>=3

97 65 39 26 16 10 3 1 0

GEP70 > 0.66

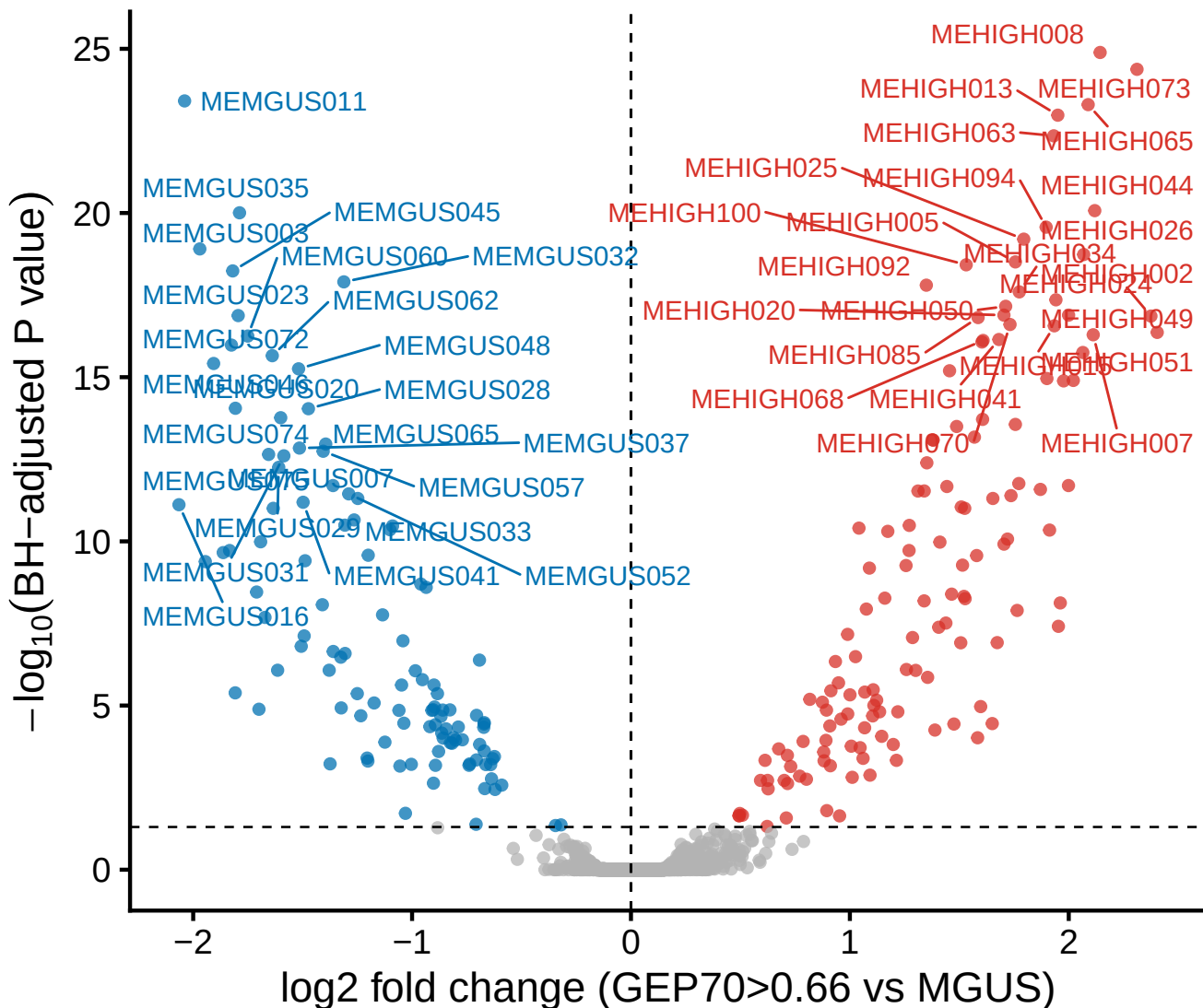
43 30 17 12 7 4 1 0 0

Selected RNBX Whole-Bone-Marrow Cohort

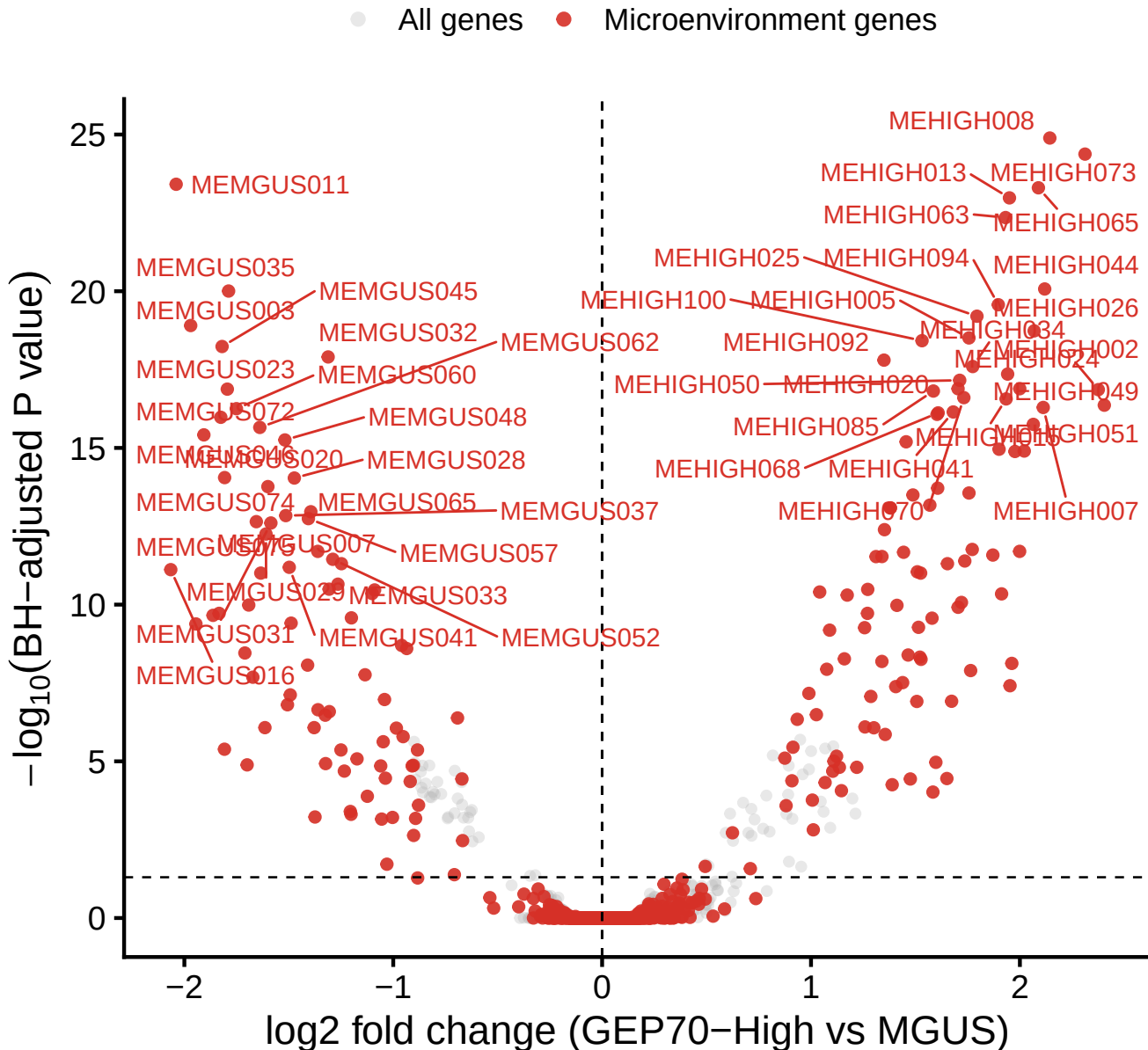


MGUS vs GEP70-High

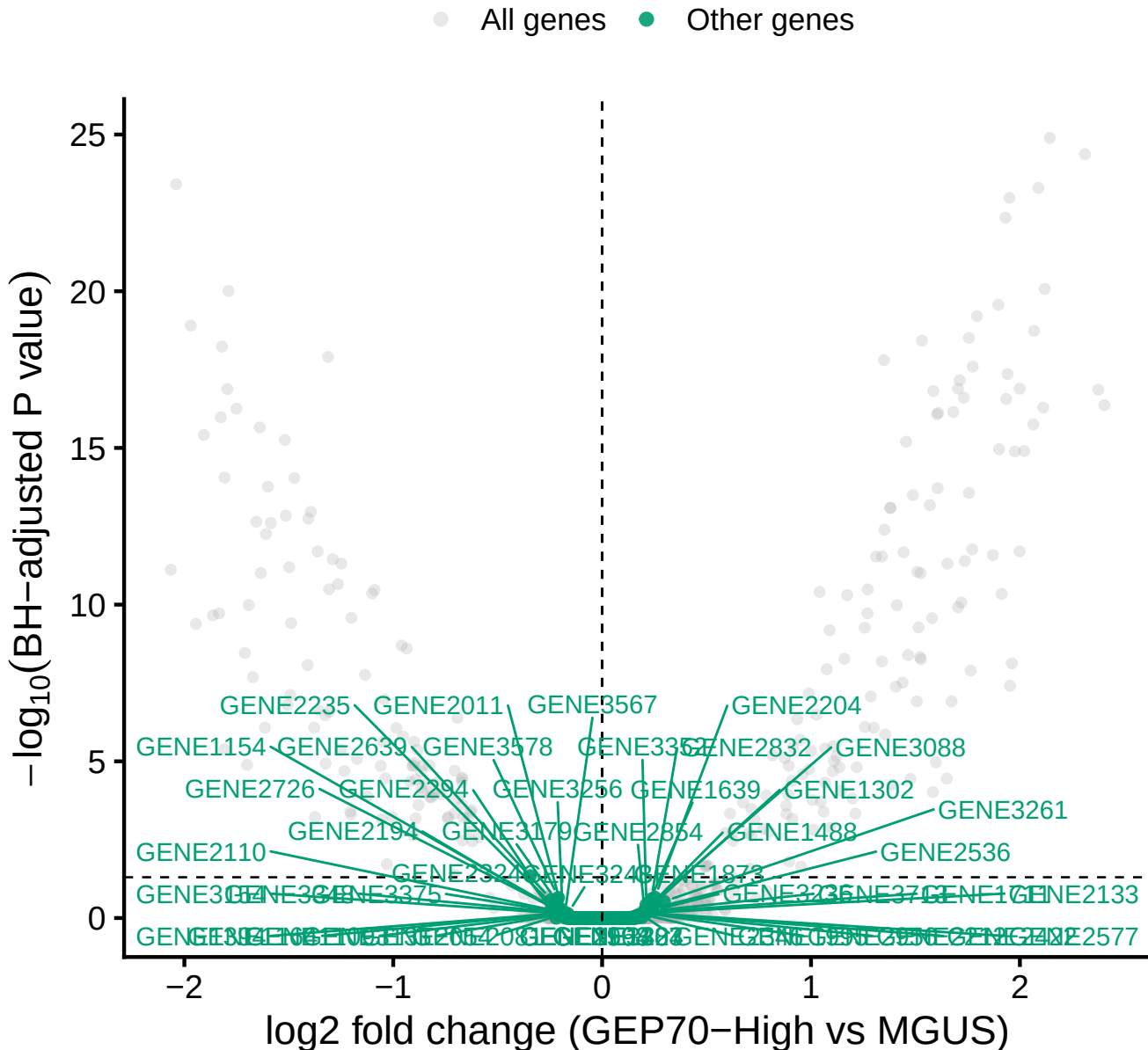
● Up in MGUS ● Not significant ● Up in GEP70>0.66



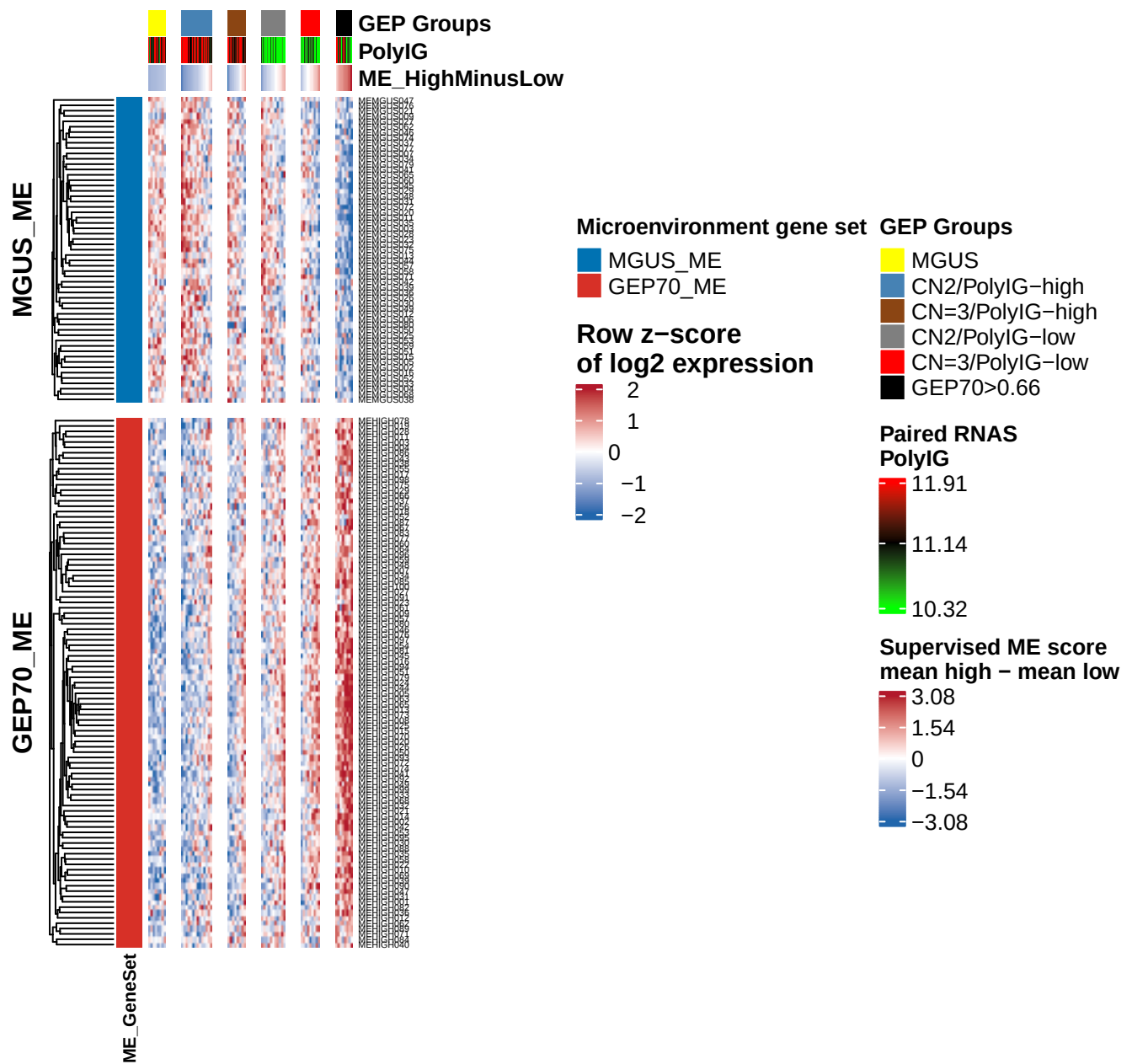
MGUS vs GEP70-High: Microenvironment



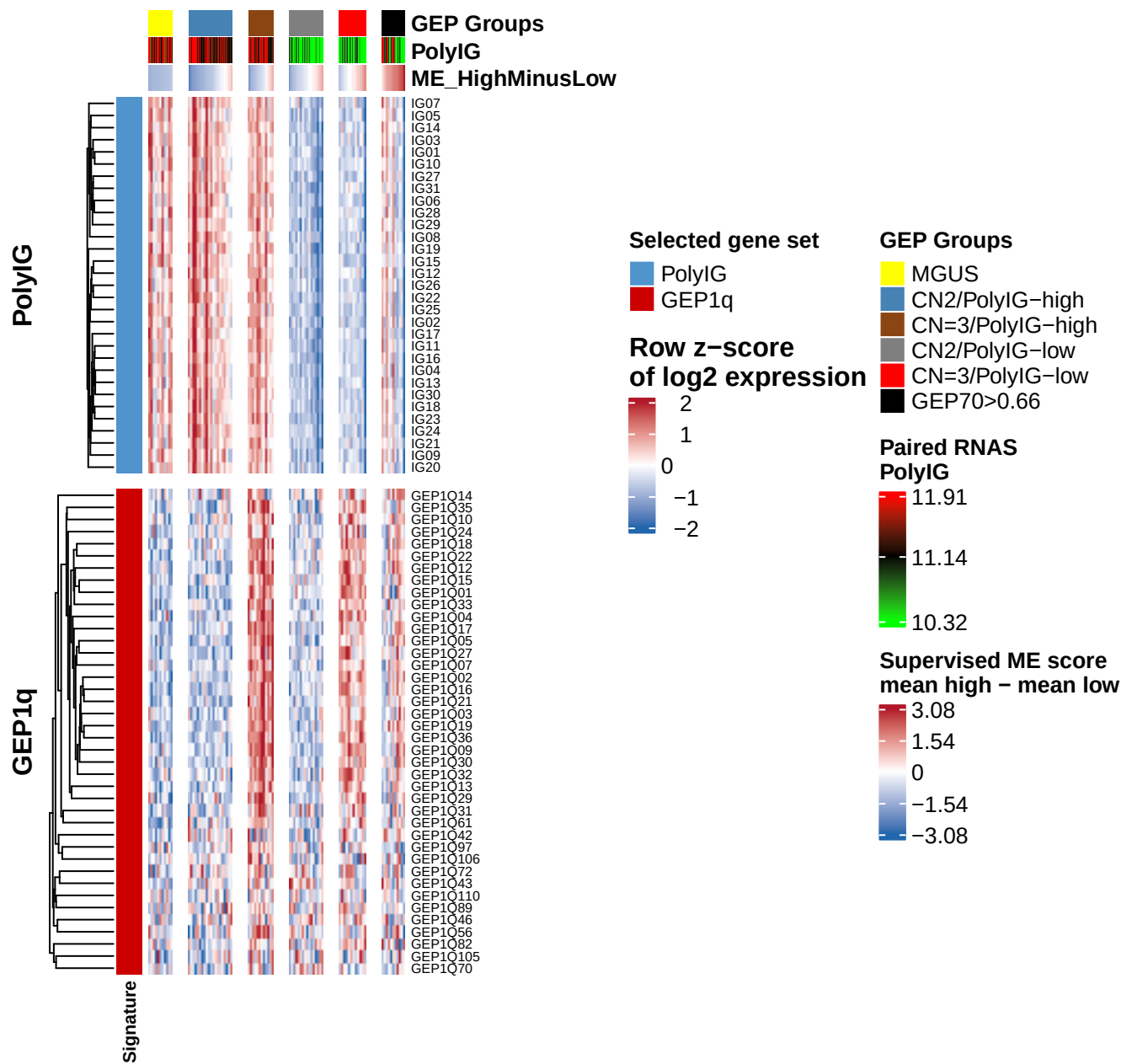
MGUS vs GEP70-High: Other genes



t Gene Signature Across MGUS and NDMM Risk Group



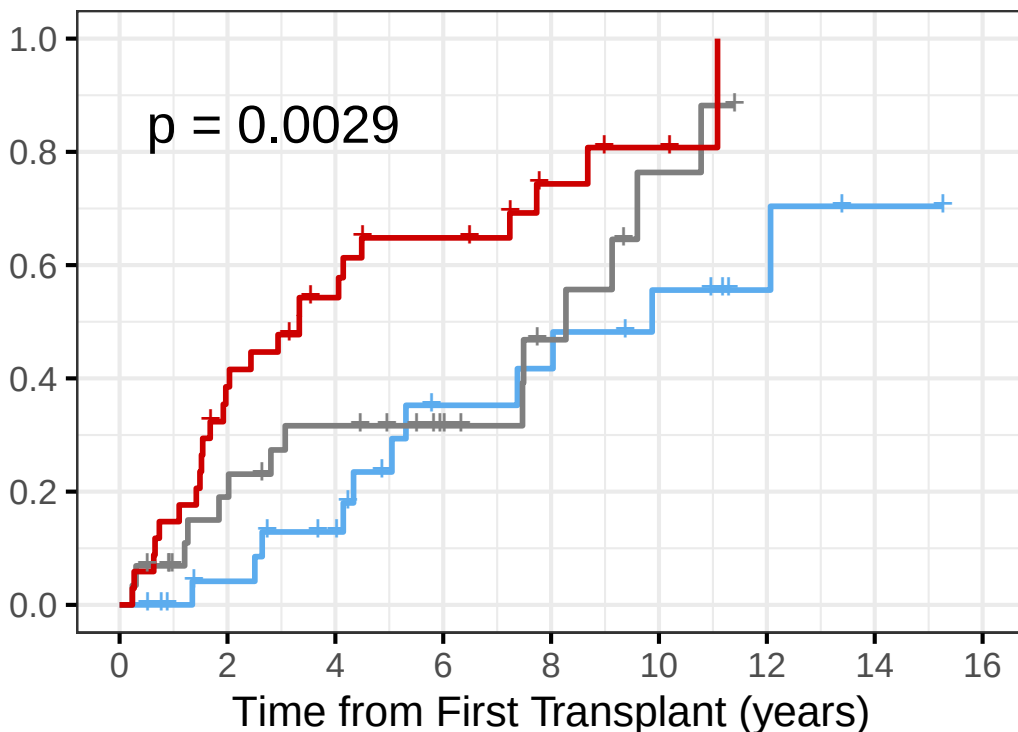
Immunoglobulin Expression Across MGUS and NDM



Time to Progression by GEP 1q CN × PolyIG (M

Standard Risk (GEP70); TT2 (n = 90)

Cumulative Progression Probability



Group + CN2/PolyIG High + CN2/PolyIG Low + CN>=3/PolyIG ALL

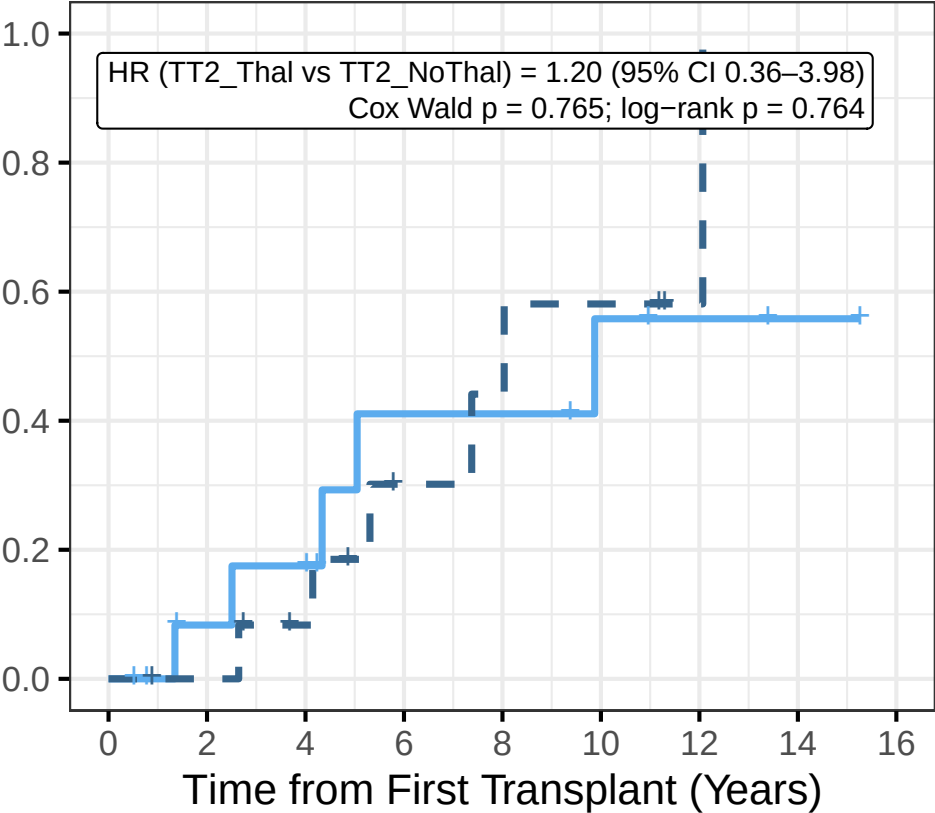
No. at risk

CN2/PolyIG High	27	22	18	10	9	6	3	1	0
CN2/PolyIG Low	29	20	16	11	6	2	0	0	0
CN>=3/PolyIG ALL	34	20	13	9	4	2	0	0	0

1q2/PolyIG–High Time to Progression in TT2 With vs With

GEP70 standard risk; n = 27; PolyIG > 11.00

Cumulative Progression Probability



strata  TT2 No Thalidomide (n = 14)  TT2 Thalidomide (n = 13)

No. at risk

TT2 No Thalidomide (n = 14)

14 10 9 5 5 3 2 1 0

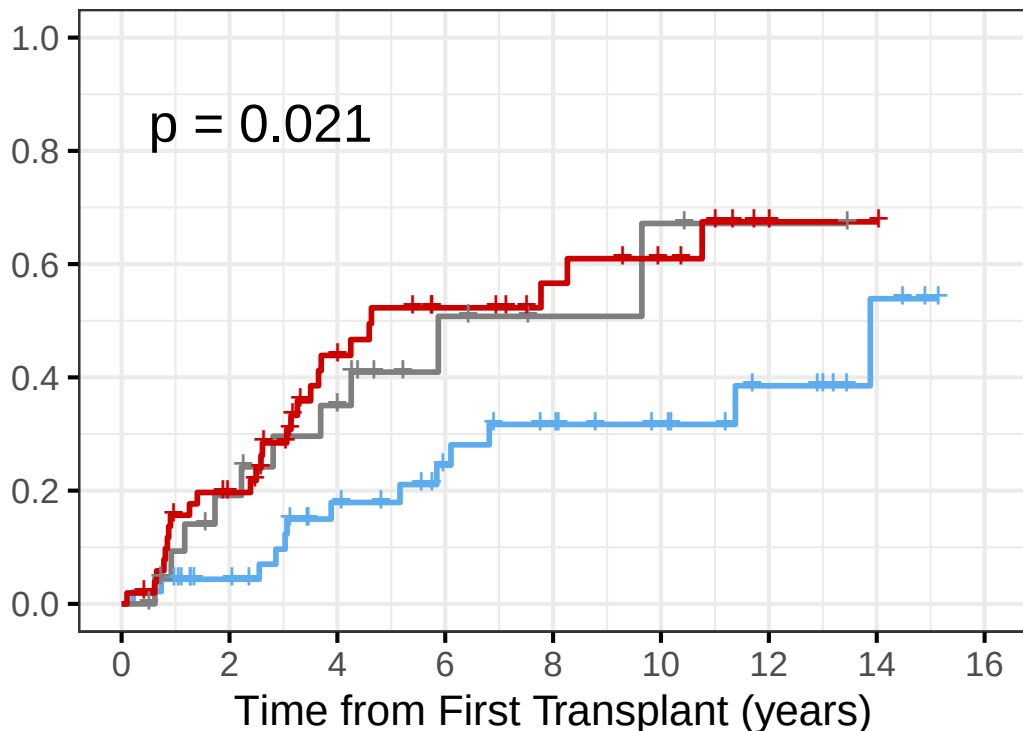
TT2 Thalidomide (n = 13)

13 12 9 5 4 3 1 0 0

Time to Progression by GEP 1q CN × PolyIG (PI)

Standard Risk (GEP70); TT3a, TT3b, TT4-S (n = 100)

Cumulative Progression Probability



Group + CN2/PolyIG High + CN2/PolyIG Low + CN≥3/PolyIG ALL

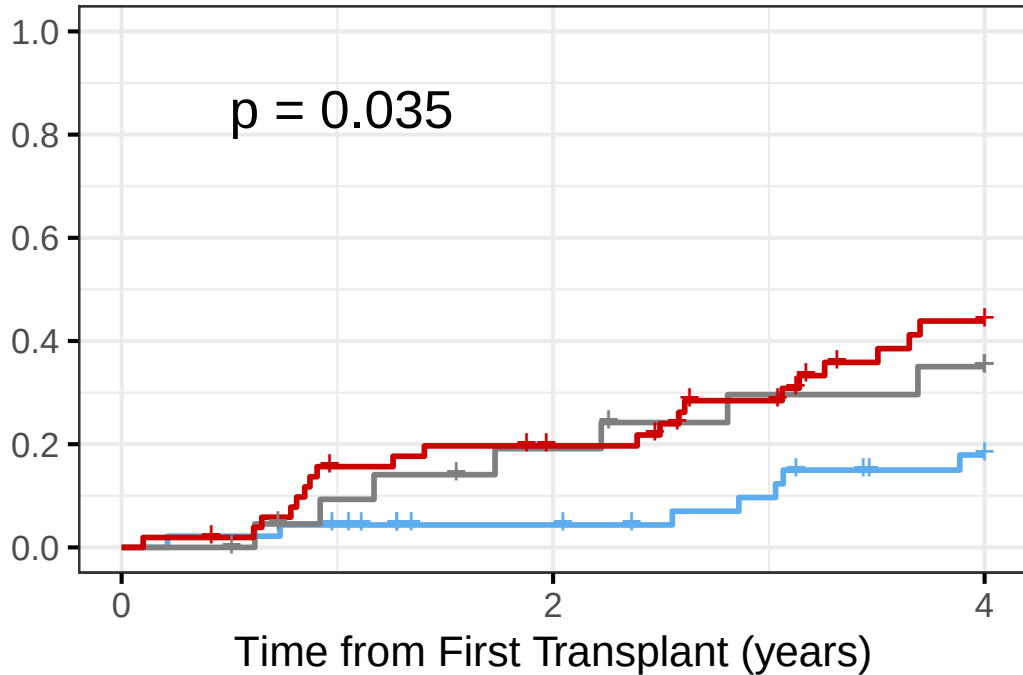
No. at risk

CN2/PolyIG High	46	38	28	21	17	13	8	3	0
CN2/PolyIG Low	23	16	11	5	3	2	1	0	0
CN≥3/PolyIG ALL	52	38	21	14	10	7	2	1	0

Time to Progression by GEP 1q CN × PolyIG (PI) Administratively Censored at 4 Years

Standard Risk (GEP70); TT3a, TT3b, TT4-S (n = 1

Cumulative Progression Probability



Group + CN2/PolyIG High + CN2/PolyIG Low + CN≥3/PolyIG ALL

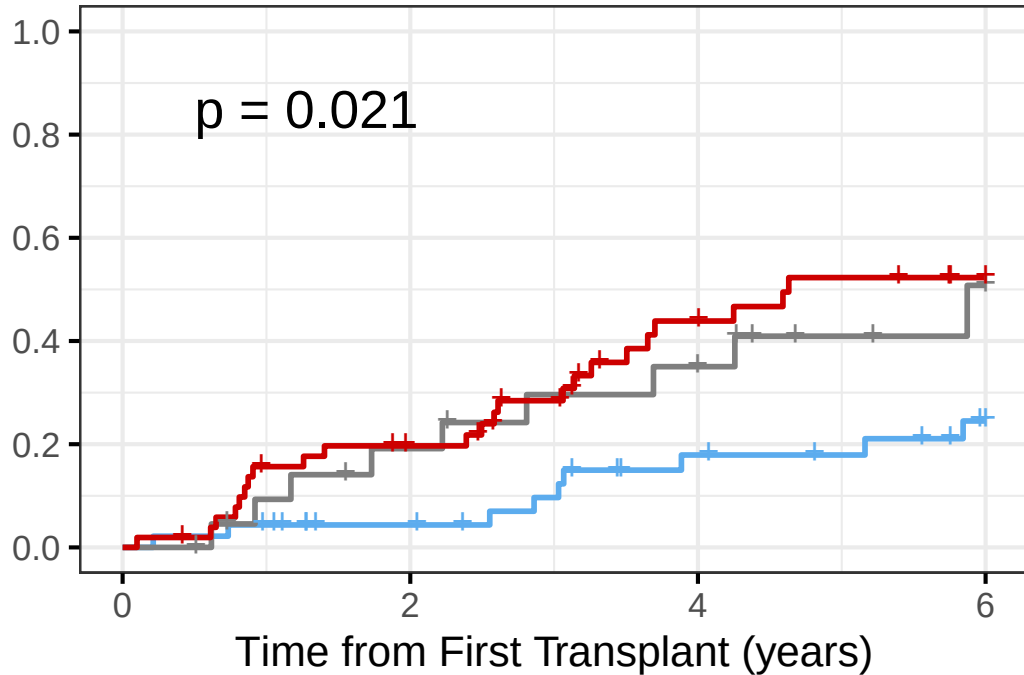
No. at risk

CN2/PolyIG High	46	38	28
CN2/PolyIG Low	23	16	11
CN≥3/PolyIG ALL	52	38	21

Time to Progression by GEP 1q CN × PolyIG (PI) Administratively Censored at 6 Years

Standard Risk (GEP70); TT3a, TT3b, TT4-S (n = 1

Cumulative Progression Probability



Group + CN2/PolyIG High + CN2/PolyIG Low + CN≥3/PolyIG ALL

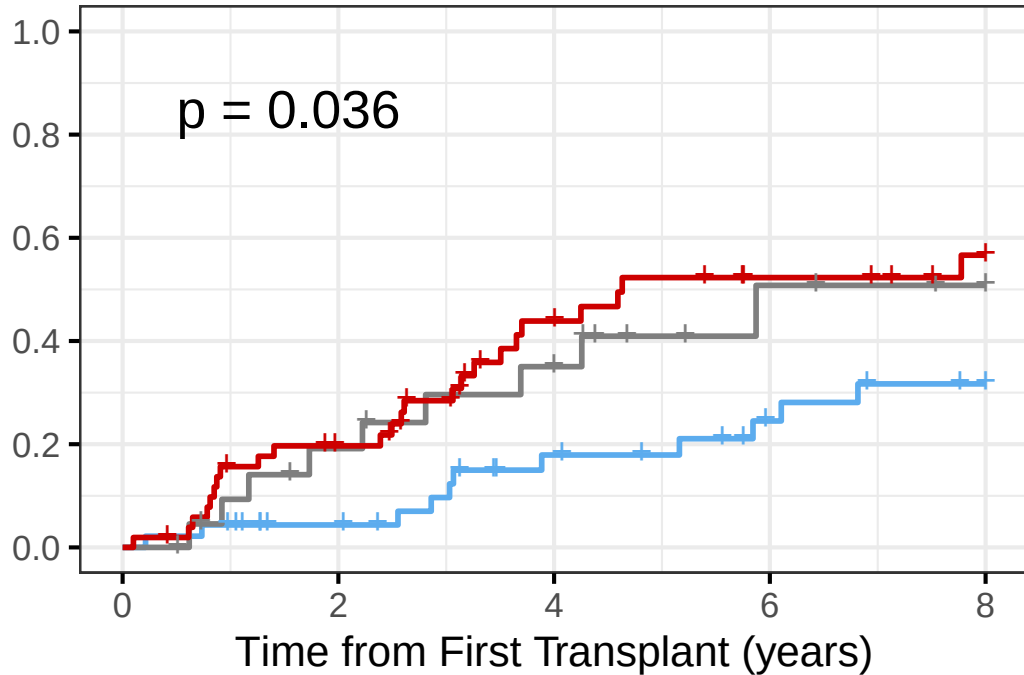
No. at risk

CN2/PolyIG High	46	38	28	21
CN2/PolyIG Low	23	16	11	5
CN≥3/PolyIG ALL	52	38	21	14

Time to Progression by GEP 1q CN × PolyIG (PI) Administratively Censored at 8 Years

Standard Risk (GEP70); TT3a, TT3b, TT4-S (n = 1

Cumulative Progression Probability



Group + CN2/PolyIG High + CN2/PolyIG Low + CN≥3/PolyIG ALL

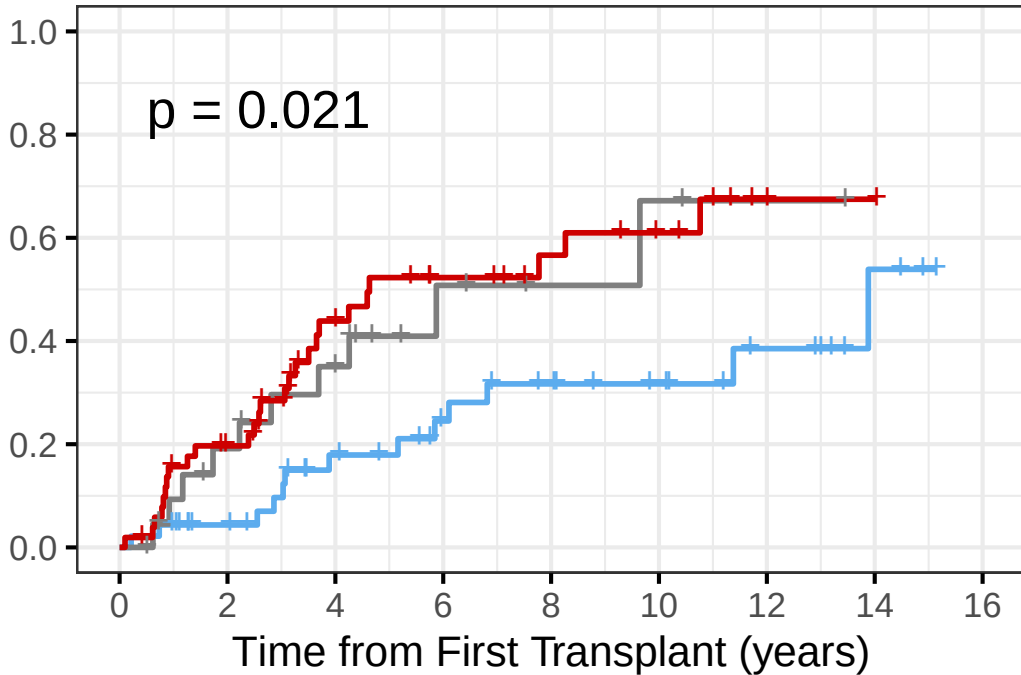
No. at risk

CN2/PolyIG High	46	38	28	21	17
CN2/PolyIG Low	23	16	11	5	3
CN≥3/PolyIG ALL	52	38	21	14	10

Time to Progression by GEP 1q CN × PolyIG (PI) Administratively Censored at Full Follow-up

Standard Risk (GEP70); TT3a, TT3b, TT4-S (n = 1

Cumulative Progression Probability



Group + CN2/PolyIG High + CN2/PolyIG Low + CN>=3/PolyIG ALL

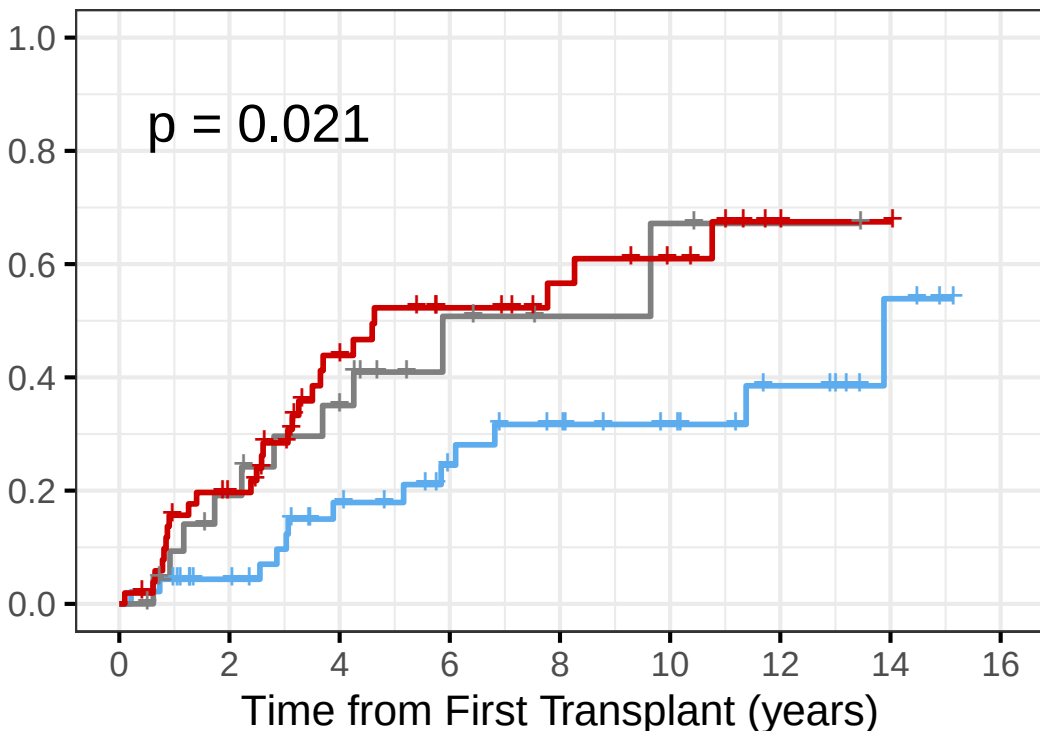
No. at risk

CN2/PolyIG High	46	38	28	21	17	13	8	3	0
CN2/PolyIG Low	23	16	11	5	3	2	1	0	0
CN>=3/PolyIG ALL	52	38	21	14	10	7	2	1	0

Time to Progression by GEP 1q CN × PolyIG (PI)

Standard Risk (GEP70); TT3a, TT3b, TT4-S (n = 1

Cumulative Progression Probability



Group CN2/PolyIG High CN2/PolyIG Low CN=3/PolyIG

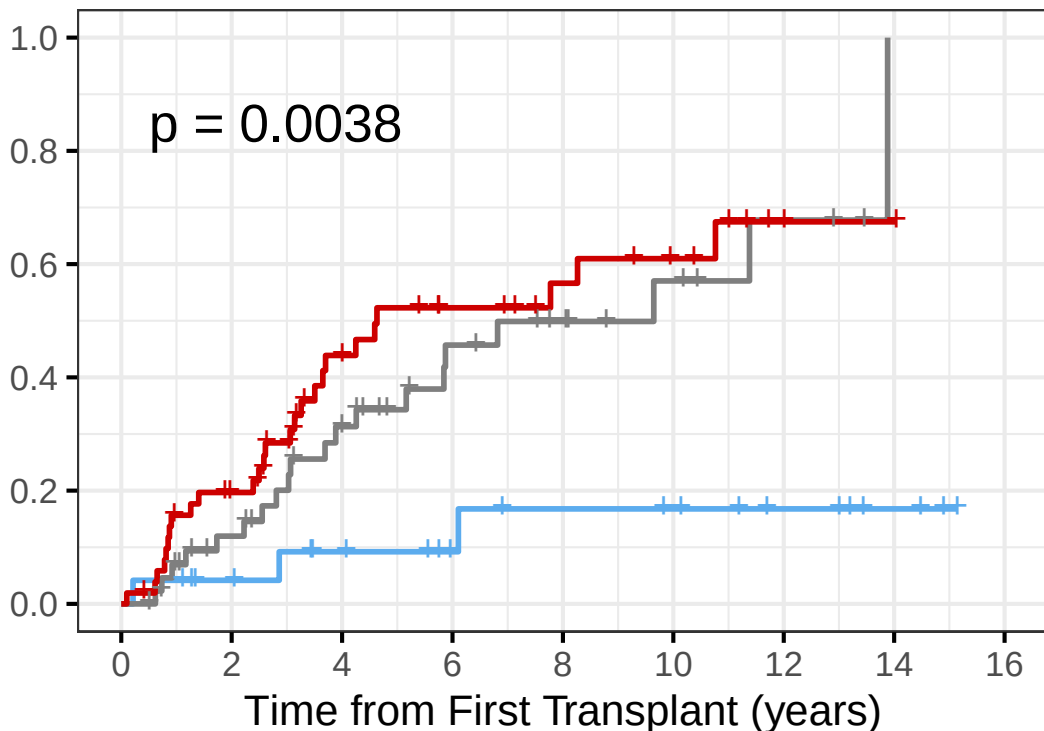
No. at risk

CN2/PolyIG High	46	38	28	21	17	13	8	3	0
CN2/PolyIG Low	23	16	11	5	3	2	1	0	0
CN=3/PolyIG ALL	52	38	21	14	10	7	2	1	0

Time to Progression by FISH 1q CN × unIGH (PI)

Standard Risk (GEP70); TT3a, TT3b, TT4-S (n = 1

Cumulative Progression Probability



Group CN2/unIGH High CN2/unIGH Low CN=3/unIGH

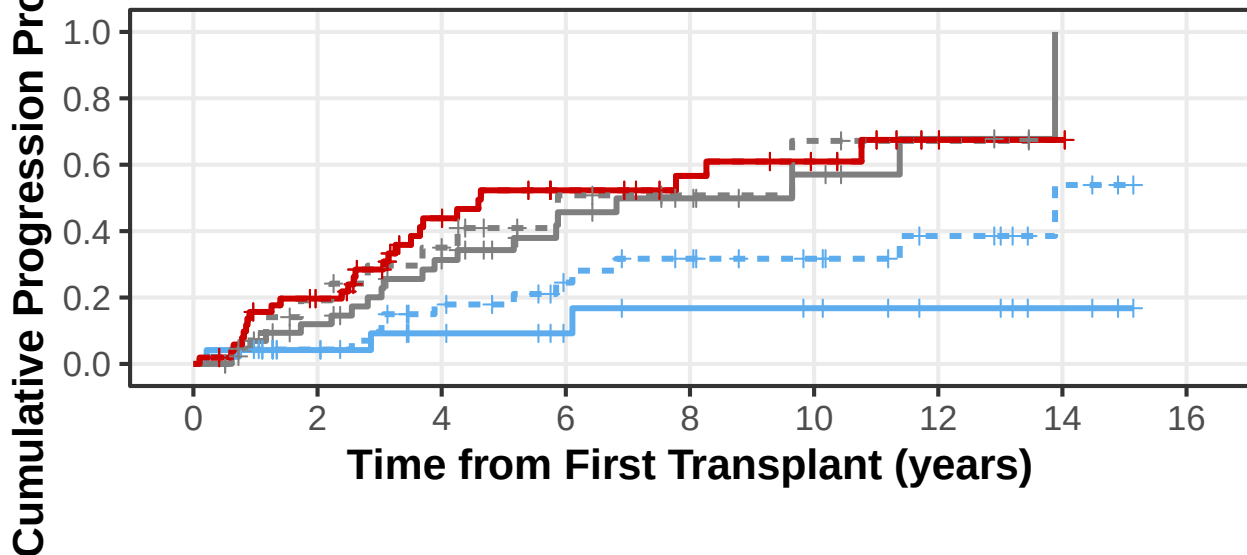
No. at risk

CN2/unIGH High	24	20	16	12	10	9	6	3	0
CN2/unIGH Low	45	34	23	14	10	6	3	0	0
CN=3/unIGH ALL	52	38	21	14	10	7	2	1	0

Clinical and GEP-Derived Risk Groups

Same matched patients (n = 121)

Clinical log-rank p = 3.8e-03; GEP log-rank p = 2.1e-02



Risk Group + Favorable + Intermediate + Adverse

Measurement — Clinical FISH 1q + unIGH - - GEP1q + PolyIG

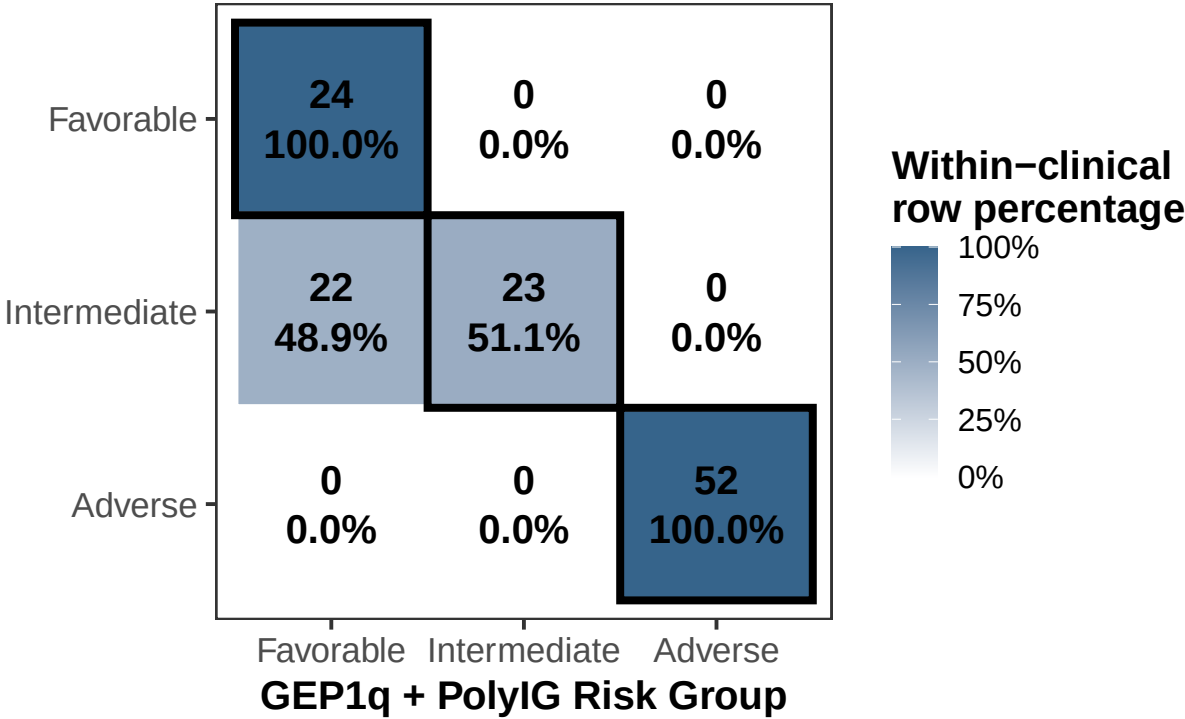
No. at risk

Favorable — Clinical	24	20	16	12	10	9	6	3	0
Favorable — GEP	46	38	28	21	17	13	8	3	0
Intermediate — Clinical	45	34	23	14	10	6	3	0	0
Intermediate — GEP	23	16	11	5	3	2	1	0	0
Adverse — Clinical	52	38	21	14	10	7	2	1	0
Adverse — GEP	0	2	4	6	8	10	12	14	16

Clinical and GEP-Derived Risk-Group Agreement

Matched patients: n = 121 | Exact agreement = 81.8% | Cohen's κ = 0.728

Clinical FISH1q + unIGH Risk Group



Cell labels show patient count and percentage within each clinical row. Black outlines mark